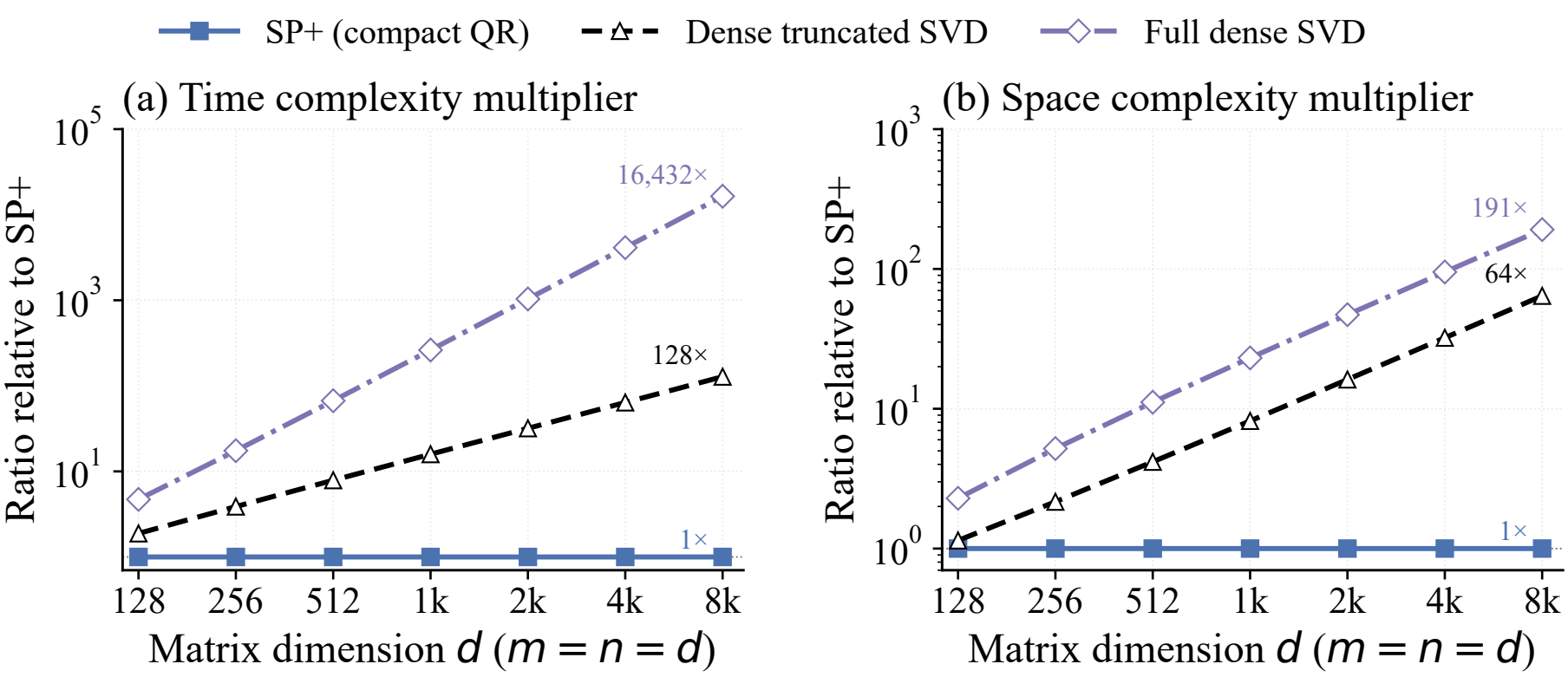
SP+ (compact QR)
Dense truncated SVD
Full dense SVD
(a) Time complexity multiplier
Ratio relative to SP+
10^5
10^3
10^1
16,432×
128×
1×
128
256
512
1k
2k
4k
8k
Matrix dimension d ($m = n = d$)
(b) Space complexity multiplier
Ratio relative to SP+
10^3
10^2
10^1
10^0
191×
64×
1×
128
256
512
1k
2k
4k
8k
Matrix dimension d ($m = n = d$)